TRANG TRUONG

(650) - 267 - 1710 | trangtruong.071105@gmail.com | LinkedIn | Portfolio | GitHub

EDUCATION

Arizona State University ***Aug 2023 – Dec 2026***

Bachelor of Science, Business Data Analytics

- **GPA:** 3.83
- **Coursework:** Business Statistics, Data Structures and Algorithms, Machine Learning, Object-Oriented Programming, Software Engineering and System Design, Social Media Mining

SKILLS

- **Programming Languages:** Python (Pandas, NumPy, Scikit-learn, PyTorch, TensorFlow), SQL, C++, C#, HTML, CSS, TypeScript
- **Machine Learning & AI:** Supervised/Deep Learning, NLP, LLMs, Retrieval-Augmented Generation (RAG), Semantic Search, Multi-Label Classification, Computer Vision (YOLOv8), Generative AI, Signal Processing, Vector Embeddings (pgvector), Agentic AI, Multimodal AI, Responsible AI (Bias Detection, Hallucination Mitigation), Video Generation (Veo 3)
- **Frameworks & Tools:** HuggingFace Transformers, OpenCV, FastAPI, Next.js, Jupyter Notebook, Git, Docker
- **Cloud & Platforms**: AWS Lambda, Databricks, Spark, Google Gemini API, Supabase, Vercel
- **Database & Querying:** SQL, MySQL, NoSQL, FAISS, PostgreSQL
- **Data Visualization:** Matplotlib, Plotly, Tableau
- **Soft Skills:** Problem Solving, Analytical Thinking, Communication, Leadership, Teamwork

EXPERIENCE

BullyBlocker ***Jul 2025 - Present***

Undergraduate Researcher

- Optimized cyberbullying detection model by integrating multiple API-based LLMs, achieving a 10-13% F1-score improvement and enabling scoring of harmful conversations.
- Developed explainable NLP systems that identified key reasons behind flagged abuse, reducing false positives by 12% and enhancing trust in moderation outcomes.
- Designed and deployed LLM-powered mediation generation models, advancing ethical AI applications for safer and more constructive online interactions.

School of Mathematical and Statistical Sciences, ASU ***Oct 2024 - Jun 2025***

Office Assistant

- Automated scheduling workflows for students, cutting processing delays by 30% and improving transparency
- Digitized math placement records, accelerating data access and reducing registration bottlenecks.

ACHIEVEMENTS

EduTag LINK| Winner – Rice Datathon 2026 ***Jan 2026***

- Designed and implemented a Hybrid Retrieval-Augmented Fine-Tuning (RAFT) pipeline using Bi-Encoder and Cross-Encoder Transformer architecture (all-mpnet-base-v2 + FAISS) to semantically align textbook content with curriculum standard codes, achieving 41.5% exact match accuracy and 0.026 Hamming Loss across 156 standard codes
- Engineered a Gemini API-powered data augmentation workflow to generate 1,000+ synthetic training samples, expanding label coverage from 76 to 156 distinct standard codes and significantly improving model performance on rare/long-tail classifications

WALL-E LINK | Winner – Vandyhacks XII ***Mar 2026***

- Architected and implemented a **central state machine** fusing multi-modal inputs - EEG jaw clench, eye gaze, and head movement - into a unified real-time control pipeline driving a 6-DOF robotic arm with no physical interaction required
- Led backend integration of three independent AI/signal processing systems (YOLOv8 object detection, Muse 2 EEG decoding, and webcam head tracking) into a cohesive assistive device, coordinating input gating logic to prevent conflicting signals across modalities
- Engineered real-time intent recognition pipeline translating raw brain and vision signals into precise robotic actions via inverse kinematics, enabling hands-free object manipulation for patients with paralysis or ALS

PROJECTS

ChronosVR - Generative AI Historical Simulation Platform LINK | ***Veo 3, Gemini Pro, PyTorch, TensorFlow, C#*** ***Dec 2025***

- Designed and implemented a C#-based multimodal GenAI pipeline that transforms unstructured historical data into interactive 3D simulations, orchestrating Gemini Pro for narrative generation and PyTorch-based agentic models to control NPC decision-making and behavior.
- Built Responsible AI grounding and evaluation layers using TensorFlow, validating generated outputs against curated historical sources, monitoring bias and uncertainty, and applying automated feedback loops to reduce hallucinations and improve output reliability.